

## CSC15S Assignment 4 – Hash and Linux File Systems II

**Total: 100pts (due date - see Blackboard)**

What to turn in: upload the following to **Assignment 4** in Blackboard.

- (1) CSC15Slab4.docx (with all questions completed)
- (2) log\_Mission2.txt (by default, it's located in ~/Downloads folder)

## Part 1. Hash [20 pts. Estimated time to complete: 20 minutes]

1. Read about the hack of Linux Mint Download Website [src: <https://blog.linuxmint.com/?p=2994>]. Can the use of MD5 (or similar checksum techniques) help prevent such events completely (given that the site that hosts the .iso image and the MD5 checksum can be hacked and modified by malicious attackers)? [5 pts]

MD5 or other checksum techniques can verify the integrity of files by comparing the calculated checksum with the expected one. However, if the server hosting both the ISO image and the checksum is compromised, attackers can alter both the image and the checksum, rendering MD5 ineffective in preventing such attacks completely. Therefore, checksums alone cannot fully prevent these events if the source is hacked.

2. Finish Reading I (in Blackboard). Describe the difference between collision resistance and preimage resistance, with no more than 50 words. [10 pts]

**Collision resistance:** It's hard to find two different inputs that produce the same hash.

**Preimage resistance:** Given a hash, it's hard to find the original input that generated the hash.

3. A SHA-1 hash is 160-bit. In theory, using the generic birthday attack, how many attempts would be needed? You can express it in the form of power of 2, e.g.,  $2^{20}$ . [5 pts]

$2^{80}$

## Part 2. Dr. Evil's Training Room. Mission 2 – The Dungeon of Linux Files.

[80 pts. Estimated time to complete: 1.5 hrs]

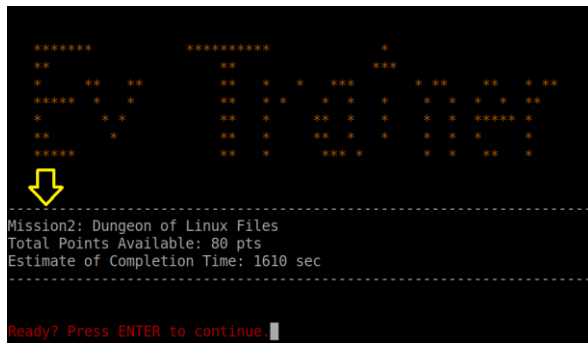
Dr. Evil has prepared some nice “appetizers” for you. It’s an automated bot grader and trainer. Let’s download it first:

- (1) Start a browser in your CSC15S VM Image and visit Blackboard class account
- (2) Navigate to Assignments -> Assignment 4
- (3) Right click on mission\_suid and download the file (make sure that it’s named as **mission\_suid** in the download dialog). You can save it to the Downloads folder.

Then you can start the program following the commands below:

```
cd ~/Downloads
chmod 755 mission_suid
sudo ./mission_suid
```

The password is “goodyear”. Please make sure you are doing the right mission. See Figure below:



As usual, you need to start another terminal to complete the training. Once you feel ready to submit, please attach log\_Mission2.txt to Blackboard.

The following are some additional hints for each of the challenges.

| Challenge        | Hint                                                                                                                                  |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1. Checksum      | Use just one command <b>md5sum *</b> to generate checksums and compare.                                                               |
| 2. Simple suid   | Use <b>sudo chmod</b> command properly                                                                                                |
| 3. File info     | If the xxd output is too long, use file redirection, e.g., “ <b>xxd abc.bin &gt; dump.txt</b> ”                                       |
| 4. sed_cat       | Option “-i” allows you to modify the input file directly. E.g., <b>sed -i “1i TEST” abc.txt</b> inserts “TEST” into line 1 of abc.txt |
| 5. grep_find     | Use “-R” for the grep command to search in nested folder.                                                                             |
| 6. chain of cmds | Review the difference between <b>{ } \;</b> and <b>{ } \+</b> first.                                                                  |